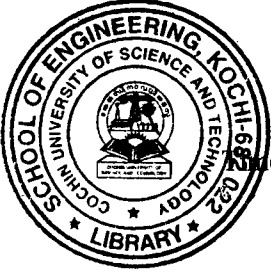


**B.Tech. Degree II Semester Regular and Supplementary/
I Semester Supplementary Examination May 2017**

**CE/EE/ME/SE GE 15-1204 A & CS/EC/IT GE 15-1104 B
BASIC ELECTRICAL ENGINEERING
(2015 Scheme)**



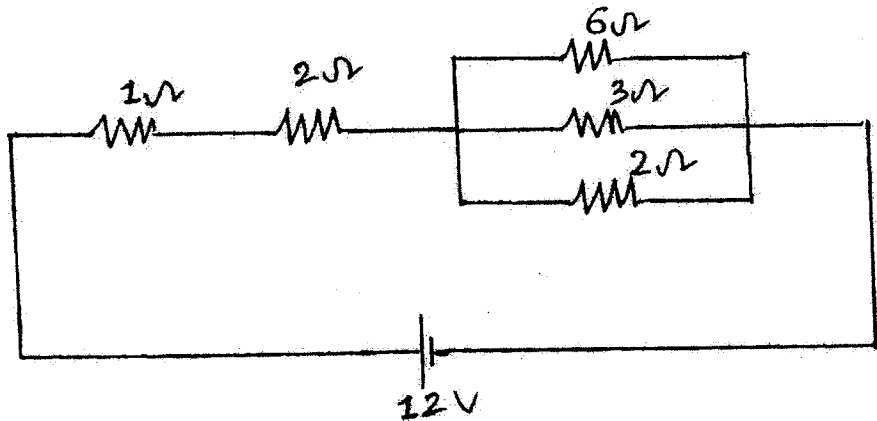
Time: 3 Hours

Maximum Marks: 60

PART A
(Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) In the circuit given below, find the current through all the resistances.

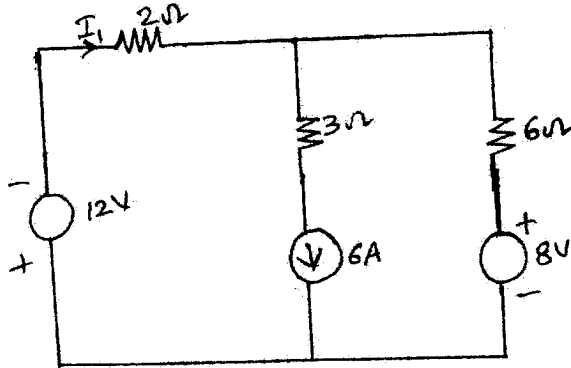


- (b) Define capacitance. Derive an expression for the total capacitance of a group of capacitors when they are connected in (i) parallel and (ii) series.
- (c) State and explain Norton's theorem.
- (d) A magnetic field is produced by a coil of 300 turns which is wound on a closed iron ring. The ring has a cross section of 20cm^2 and mean length of 120cm. The permeability of iron is 800. If the current in the coil is 10A, find the energy stored in the magnetic field.
- (e) Two similar coils connected in series gave a total inductance of 600mH and when one coil is reversed, the total inductance is 300mH. Determine the mutual inductance between the coils and coefficient of coupling.
- (f) Define the following terms (i) cycle (ii) frequency (iii) wave length and (iv) velocity.
- (g) The apparent power drawn by an ac circuit is 10KVA and active power is 8kW. What is the reactive power in the circuit? What is the power factor of the circuit?
- (h) Derive the relation between line and phase values of voltage and current in a balanced delta connected system with the help of phasor diagram.
- (i) Draw the circuit diagrams of the different self excited dc generators.
- (j) Draw the diagram of a hydro electric power station and state the advantages and disadvantages.

PART B

(4 × 10 = 40)

- II. (a) State and explain Thevenin's theorem. (5)
 (b) Determine the current I_1 in the circuit by superposition theorem. (5)



OR

- III. (a) A 5MF capacitor is charged to 100V and a 10MF capacitor is charged to 250V. They are connected in parallel with terminals of like polarity together. Calculate the final charge on each capacitor and the potential difference across them. (7)
 (b) State and explain Ohms law. What are its limitations? (3)
- IV. (a) Two coils A of 5000 turns and B of 3000 turns lie in parallel planes. A current of 6A in coil A produces a flux of 0.1mwb. If 60% of the flux produced by coil A links with the turns of coil B, calculate the emf induced in coil B when current in coil A changes from 5A to -5A in 0.01second. (6)
 (b) A conductor of length 100cm moves at right angles to a uniform magnetic field of flux density of 1.5wb/m² with a velocity of 50m/s. Calculate the emf induced in it. Also find the value of emf induced when the conductor moves at an angle of 30° to the direction of field. (4)

OR

- V. Explain the construction and working of an induction type single phase energy meter. (10)
- VI. (a) What is meant by rms value of an alternating current? Calculate its value for a sinusoidal current wave. (6)
 (b) Prove that the power drawn by a pure capacitor over a complete cycle is zero. (4)

OR

- VII. Two impedance $Z_1 = (6 + j8)\Omega$ and $Z_2 = (8 - j6)\Omega$ are connected in parallel across 100V supply. Determine: (10)
 (i) The power factor of each branch.
 (ii) Overall current and power factor.
 (iii) Power consumed by each branch and total power.

- VIII. (a) Explain the principle of operation of 3 phase induction motor. (6)
 (b) What are the applications of DC series motor? (4)

OR

- IX. With the help of a neat diagram explain the operation of diesel power station. (10)